

KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology (NUIST)

Nanjing, China

B.S. in Atmospheric Science (Hydrometeorology)

Sep 2023 - Jul 2027 (Expected)

- Weighted average score: 85.53/100 | GPA: 3.57/4.0 | Class ranking: 6/51 (**top 12%**)
- Research affiliate since 2023, Key Laboratory for Hydrometeorological Hazards, Ministry of Water Resources.
- Relevant coursework: Probability & Statistics, Remote Sensing for Hydrometeorology, Hydrometeorology Modeling, Numerical Weather Prediction, Python Programming.

RESEARCH INTERESTS

Remote Sensing · Hydroclimate Extremes · Land–Atmosphere Interactions · Climate Risk Analysis

PUBLICATIONS & MANUSCRIPTS

- [1] Deng, K., Huang, Z.* (2026). **Area Expansion Dominates Rising Global Three-Dimensional Terrestrial Water Storage Drought Severity.** (Manuscript in preparation)
- [2] Deng, K., Huang, Z.*, Li, H., Wu, J. (2026). **Global Unequal Exposure to Intensifying Hydrological Droughts.** *Journal of Hydrology*. (Revised manuscript under review, R3 submitted)
- [3] Wang, J.*, Deng, K. (2024). **A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean.** *STECEP*. <https://doi.org/10.61173/xnag2k18>

RESEARCH EXPERIENCE

Mitacs Globalink Research

Charlottetown, PE, Canada

Research intern | Advisor: Prof. Xander Wang (University of Prince Edward Island)

Jul 2026 - Present

- Fully funded by Mitacs Globalink and Chinese Government Scholarship (CSC) (≤ 260 awardees nationwide). Conducting land–atmosphere coupling diagnostics and GRACE-based groundwater trend analysis for Prince Edward Island.
- Comparing satellite TWS estimates with local well measurements to assess freshwater sustainability in PEI.

Global 3D Hydrological Drought Tracking & Mechanism Attribution

Sep 2025 - Present

Independent research | Advisor: Prof. Zhongwei Huang (NUIST)

- Developed a three-dimensional event-tracking framework for 23 years of GRACE/GRACE-FO terrestrial water storage observations, identifying 672 drought events and their splitting–merging lineages.
- Designed an evaporative contribution diagnostic to distinguish precipitation deficit and evapotranspiration influenced droughts and assess how spatial expansion contributes to increasing event severity.
- Principal Investigator of the Provincial Undergraduate Training Program on Innovation and Entrepreneurship.
- Refined the research design and extensions through discussions with Prof. Hui Lu at the 2026 Tsinghua DESS Forum.

Global Unequal Exposure to Intensifying Hydrological Droughts

Jun 2025 - Aug 2026

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

- Applied EEMD to GRACE/GRACE-FO CSR RL06 mascon observations and derived event-based drought metrics.
- Integrated GPW population grids to quantify a global increase of 38.76 million person-exposures yr^{-1} and decomposed the increase into hydroclimatic conditions (73.2%), population growth (5.3%), and their interaction (21.5%).
- Led the complete research workflow, from study design and global data processing to manuscript preparation and responses to reviewers; revised manuscript under review at *Journal of Hydrology* (R3 submitted).

Fluctuation Theorem & TCWV Asymmetry

Sep 2024 - Sep 2025

Independent research | Advisor: Prof. Jun Yin (NUIST)

- Extended nonequilibrium fluctuation theorem (Yin et al., 2024, *J. Climate*) from surface temperature to TCWV using ERA5; quantified PDF asymmetry A_{asym} across $\tau = 1\text{--}20$ yr smoothing timescales.

- Validated SST-TCWV Clausius-Clapeyron scaling ($R^2 = 0.95$, 7%/K); migrated tail-rescaling framework to precipitation, demonstrating universality. Produced manuscript draft in Typst and LaTeX.
- Explored extensions to precipitation through research discussions with Prof. Lamberto Rondoni (Politecnico di Torino).

WRF-based Precipitation Sensitivity over Jing-Jin-Ji

Mar 2025 - Jul 2025

Independent research | Advisor: Dr. Mengya Wang (NUIST)

- Configured WRF/WPS with 3 nested domains on Linux HPC; conducted 16 sensitivity experiments perturbing soil moisture ($\pm 10\%$ to $\pm 50\%$) and urban LULC to simulate rapid regional urbanization.
- Evaluated precipitation responses via spatial correlation, RMSE, and threat scores; identified optimal cumulus and microphysics parameterization schemes minimizing simulated biases.

Precipitation-Runoff Prediction in the Yiluo River Basin

Jun 2024 - Mar 2025

Research assistant | Advisor: Prof. Xing Yuan (Institute of Atmospheric Physics, Chinese Academy of Sciences)

- Applied EOF to monthly precipitation interannual increments, extracting 3 PCs explaining $>90\%$ variance; screened key hydroclimatic predictors via correlation analysis and stepwise regression.
- Built BiLSTM + interannual increment hybrid model: TCC improved 16%, RMSE reduced 20% vs. statistical downscaling; CSSPv2 basin runoff forecasts achieved $NSE = 0.65$ for monthly streamflow.

Contrasting SST Trends in the Southern Pacific Ocean

Mar 2024 - Sep 2024

Student researcher | Advisor: Prof. George Tselioudis (Columbia University / NASA GISS)

- Used Python to analyze 29-year multi-source satellite/reanalysis datasets (Copernicus SST, DUACS SSH, CERES, NCEP, NSIDC SIC) to characterize SST trends across the Southern Pacific (80°W - 180° , 70°S - 30°S).
- Derived Ekman transport (U_E, V_E, W_E) from wind stress to attribute the SST dipole; diagnosed a sea ice radiative positive feedback cycle. Published as co-author in *STECEP* (2024).

INTERNSHIP

Leizhou Meteorological Bureau, Guangdong

Jul 2025 - Aug 2025

Intern | Forecast Division

Zhanjiang, China

- Participated in frontline monitoring/warning for Typhoon Danas (2025) and Typhoon Wipha (2025), tracking intensification and landfall trajectories; conducted multi-model NWP comparison (ECMWF, CMA, NCEP, TRAMS).
- Calibrated field hydrologic equipment (groundwater wells, flux towers, snow sensors) utilizing telemetry systems, ensuring continuous environmental data collection for disaster forecasting.

SELECTED AWARDS & HONORS

Provincial Undergraduate Training Program: Principal Investigator (Provincial-level project)	May 2026
Chinese Government Scholarship: Chinese Scholarship Council (≤ 260 awardees nationwide)	Jan 2026
Mitacs Globalink Scholarship: Mitacs, Canadian national research organization	Jan 2026
Second Class Academic Scholarship: NUIST (top 20%)	Nov 2024 & Nov 2025
Merit Student: NUIST (top 15%)	Dec 2024

ADDITIONAL INFORMATION

Programming: Python (5 yrs+), Shell Script (5 yrs+), Fortran (2 yrs), Typst/LaTeX (3 yrs)

Tools: Google Earth Engine, Linux (HPC), Git, WRF, MATLAB, TensorFlow, PyTorch, ArcGIS

Languages: Chinese (native), English (proficient), French (basic)

Activities: Selected as one of 70 nationwide participants to deliver an English oral presentation at the 2026 National Outstanding Undergraduate Forum, Department of Earth System Science, Tsinghua University; competitive swimmer for 10 years; first-place in the NUIST departmental medley relay (2025); Maintainer of a public academic website (othodeng.github.io) sharing study notes from 10+ courses.